

An AI agent in the report editor

on an open protocol — the ACP-Rad demo

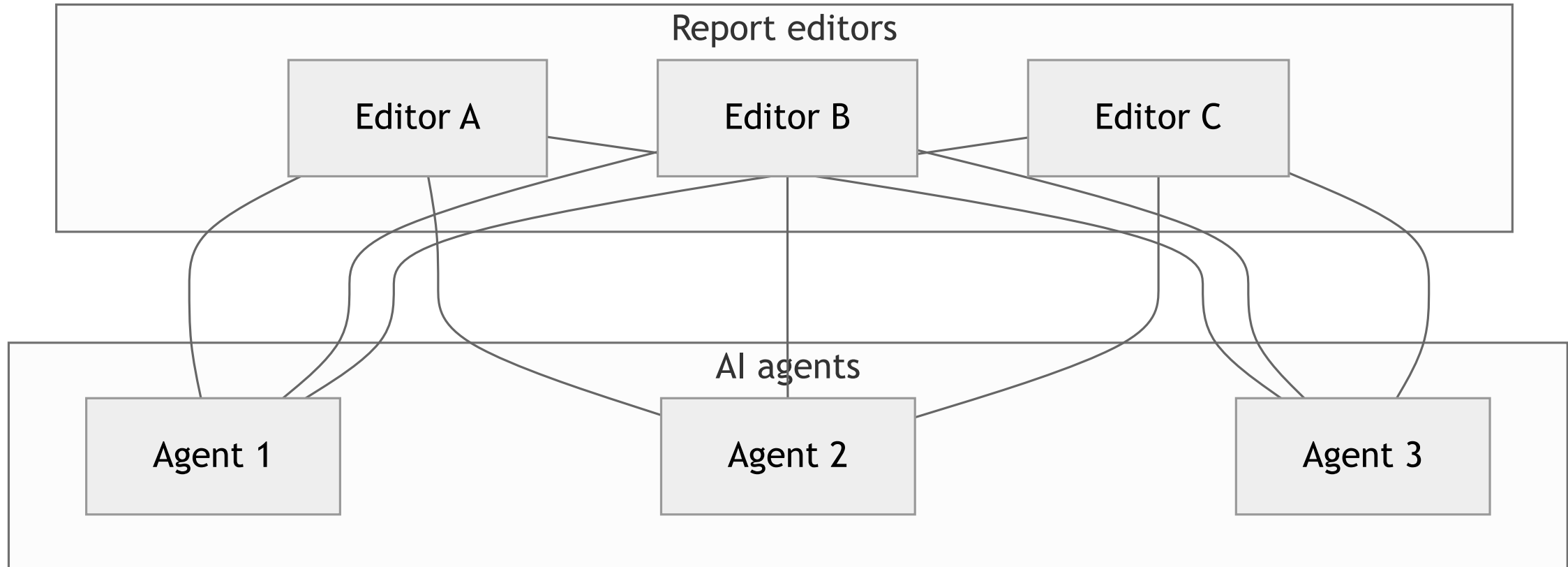
Kittipos Sirivongrungson

30 August 2026

Why

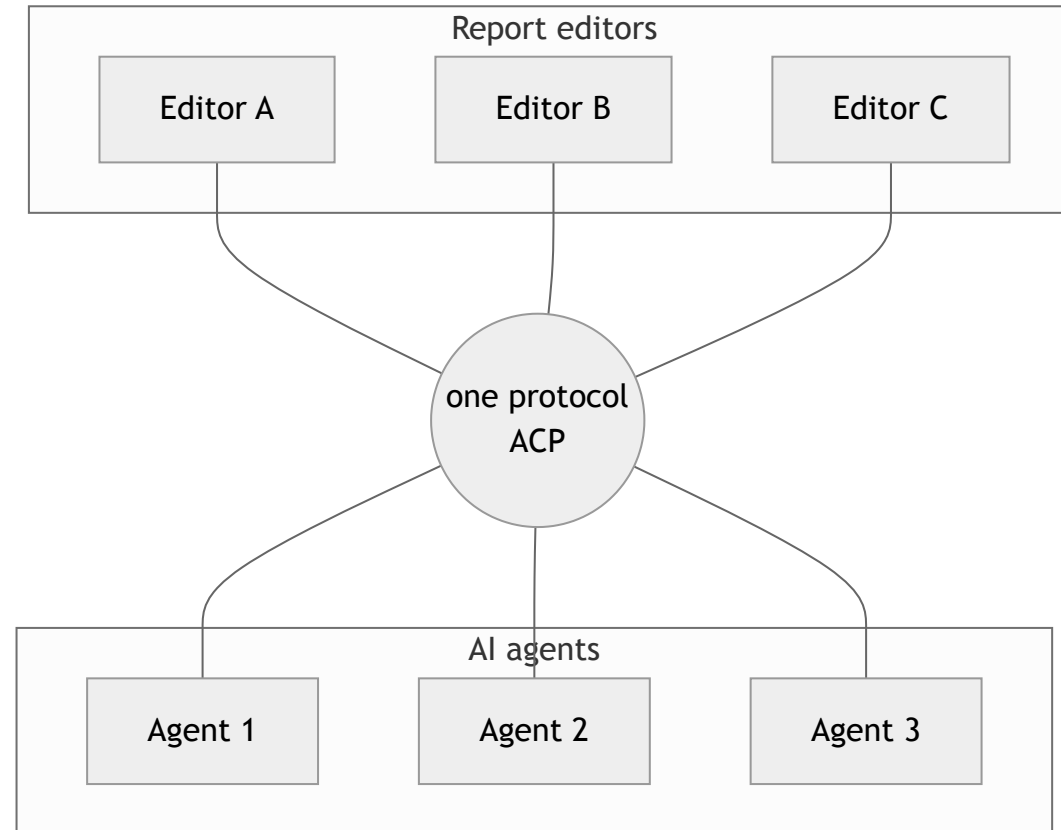
The sidebar problem, and a different way to plug an agent in.

Every AI vendor builds its own sidebar



M editors × N agents = M×N integrations. Each one decides *for itself* what the AI may write into your report.

The agent as a peripheral



Like a modality on a DICOM network: the editor speaks the protocol once, and any agent that speaks it plugs in. $M + N$.

Ground rules

The human gate

Nothing enters the report except by your act — typing, accepting a change, running a command.

The agent reads and **proposes**. It never writes on its own.

Every step is recorded — what it read, what it proposed, what you decided.

This is a technology demo.

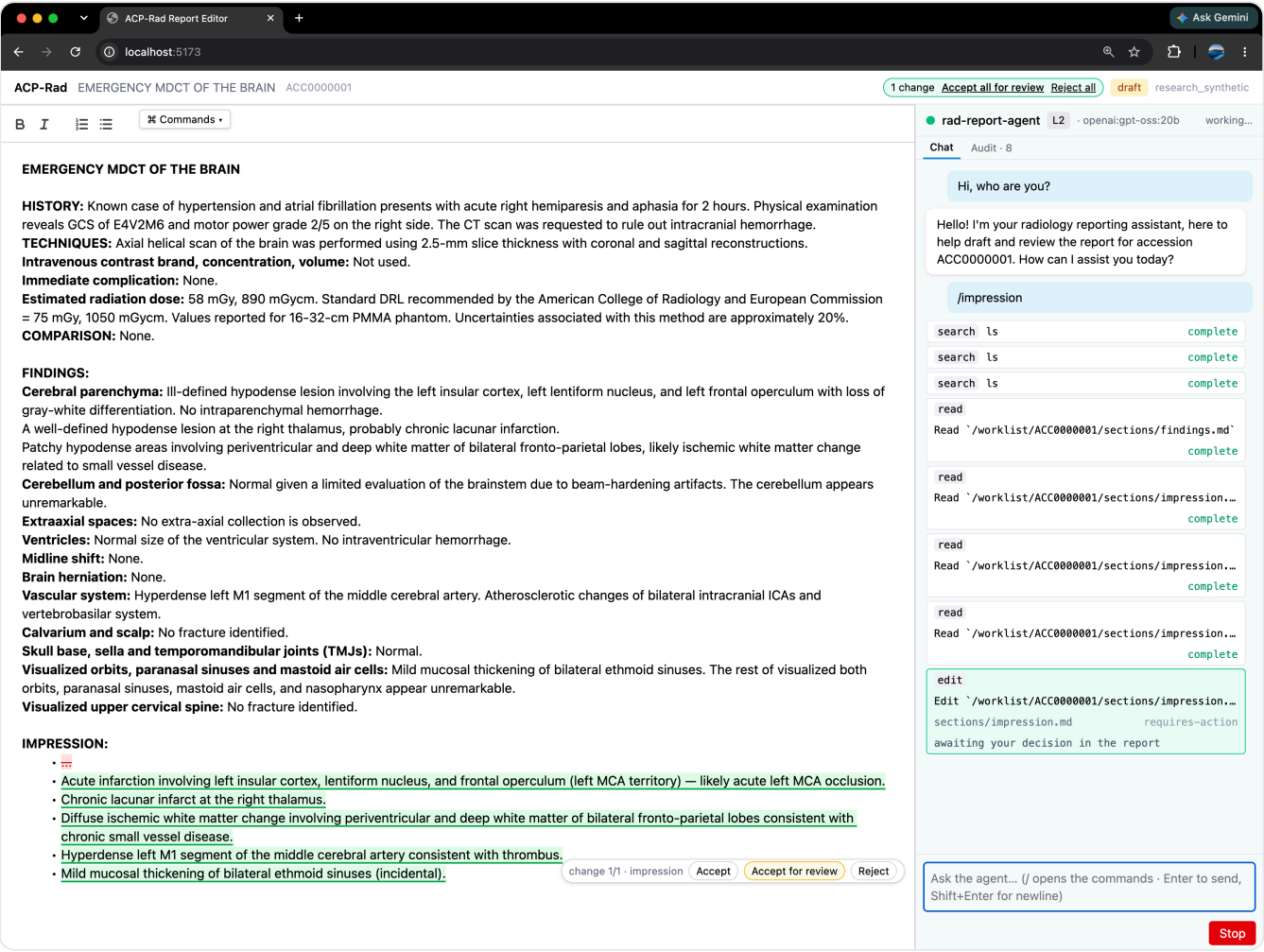
Not for clinical use. Not a medical device.

All patients, reports and priors are synthetic.

The demo

Eight moments from a working prototype. Same editor, same agent, same protocol.

The screen



Report on the left — the only text that exists. Agent on the right — Chat and Audit tabs, model selectable. Status and sign-off top right.

Start a study — no AI needed

SUGGESTED

`/template [template id]` Scaffold the house template for this study

`/short-prelim [brain | chest | body]` Issue a short prelim — the reg...

EDITOR

`/template [template id]` Scaffold the house template for this study

`/short-prelim [brain | chest | body]` Issue a short prelim — the reg...

`/er-reviewed` Attending has reviewed this preliminary report (impressi...

`/er-not-reviewed` Not yet reviewed by the attending (impression head)

`/discuss-with-dr` Record who the findings were discussed with (repo...

SKILLS

`/compare [prior accession]` Compare with the patient's priors — fill th...

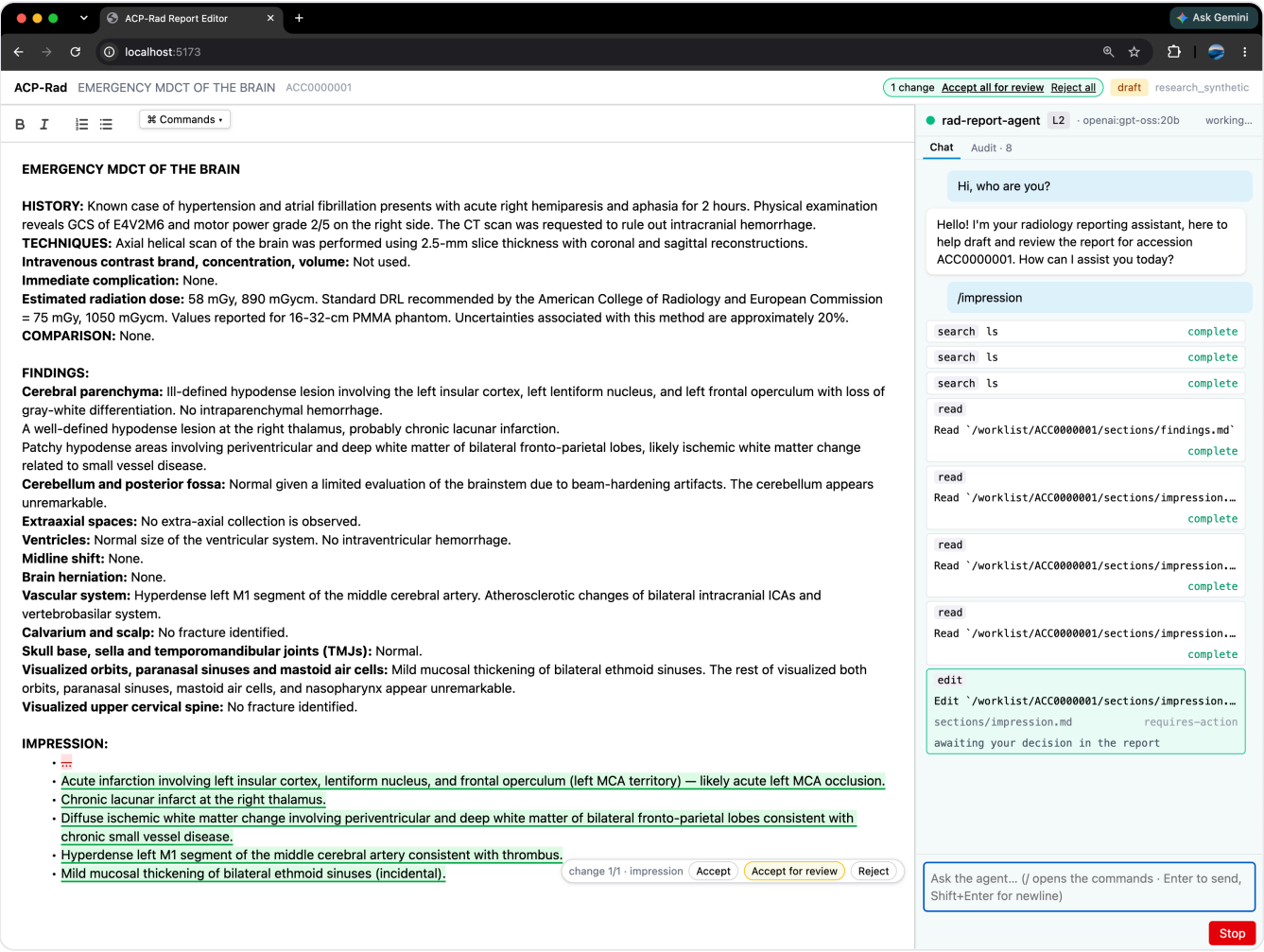
`/impression` Draft the IMPRESSION items from the FINDINGS

`/` opens one menu, three groups:

- **Editor** — instant, deterministic: `/template` scaffolds the house template; `/short-prelim` issues a short preliminary; `/er-reviewed`, `/discuss-with-dr` drop the standard lines.
- **Skills** — the agent's: `/impression` · `/compare` · `/proofread` · `/qa`.
- **Suggested** — what fits this study right now.

The boring parts stay boring. Only the *Skills* group involves an AI.

/impression — the agent proposes



The agent read FINDINGS and proposed IMPRESSION items as tracked changes. Per change: **Accept · Accept for review · Reject**.

You keep typing

IMPRESSION:

- Acute infarction in the left MCA territory — likely M1 occlusion.
- Chronic lacunar infarct at the right thalamus.
- Mild mucosal thickening of bilateral ethmoid sinuses.

green — proposed, pending your decision

amber — accepted *for review*; clears the moment you edit the line, or all at once

plain — yours

A proposal is an overlay anchored to the lines it replaces. Your typing never blocks and is never overwritten; if you change those lines first, the proposal conflicts and the agent re-reads.

Priors and polish

SKILL	READS	PROPOSES
/compare	the patient's prior reports (any modality, overlapping anatomy)	a true COMPARISON line and interval-change wording
/proofread	the section, house style	wording fixes; laterality / size / count consistency between FINDINGS and IMPRESSION
/qa	the whole report	no edit — flags only (next slide)

Priors are read-only to the agent. Every skill ends in the same place: tracked changes you decide on.

/qa — the discrepancy

ACP-RadEMERGENCY MDCT OF THE WHOLE ABDOMEN · ACC0000031residentattendingEMERGENCY MDCT OF THE WHOLE ABDOMENACC00000311 flagdraftSign offresearch_synthetic

The agent raised 1 flag — review them in the sidebar before you sign off.

B I ≡ ≡ ⚙️ Commands

EMERGENCY MDCT OF THE WHOLE ABDOMEN

HISTORY: Right flank pain for 1 day with microscopic hematuria. The CT scan was requested to rule out ureteric stone.

TECHNIQUE: MDCT axial scan with 3 mm slice thickness, coronal and sagittal reconstruction.

The scan phases include: plain, late arterial, venous, 3-minutes delayed.

Intravenous contrast brand, concentration, volume: Iohexol 350 mgI/ml; 80 ml.

Other route(s) of contrast: None

Immediate complication: None.

Estimated radiation dose: 14 mGy, 690 mGycm.

COMPARISON: None.

FINDINGS:

Liver and hepatic vasculature: Normal size and parenchymal attenuation. A 1-cm simple cyst at segment VI. Patent IVC, hepatic veins and portal veins.

Bile ducts and gallbladder: Partially-distended gallbladder without calcified gallstone. No biliary ductal dilatation.

Spleen: No splenomegaly. No focal lesion.

Pancreas: No focal mass or ductal dilatation.

Adrenals: No nodule.

Kidneys and ureters: A 6-mm calculus at the right ureteropelvic junction with mild right hydronephrosis and perinephric fat stranding. Normal size, parenchymal enhancement and excretory function of both kidneys. No other stone or mass. Normal left kidney and ureter.

Urinary bladder: Distended urinary bladder. No stone or mass.

Reproductive (Pelvic) organs: Normal size of uterus. Unremarkable both ovaries and adnexas.

Bowel and mesentery: No abnormal bowel wall thickening. No dilatation.

Lymph node: No definite intraabdominal lymphadenopathy.

Peritoneum, retroperitoneum and abdominal wall: No free fluid or free air.

Vessels: Mild atherosclerotic change of the aorta.

Bony structure: Degenerative change of spine. No suspicious lytic or blastic lesion.

Visualized lower thorax: No consolidation, nodule, mass, or tree-in-bud appearance. No pleural effusion. Normal heart size without pericardial effusion. Unremarkable both imaged breasts.

IMPRESSION:

- A 6-mm left ureteropelvic junction stone with mild hydronephrosis, likely the cause of the symptoms.
- Small simple hepatic cyst.

rad-report-agent L2 · openai:gpt-5.6-terra ready

ChatAudit · 9

discrepancysections/impression.md · line 2

FINDINGS describes a right ureteropelvic junction calculus, whereas the IMPRESSION identifies the calculus as left-sided.

Acknowledge

/qa

searchlscomplete

searchlscomplete

readRead `/worklist/ACC0000031/report.md`complete

readRead `/worklist/ACC0000031/sections/findings.md`complete

readRead `/worklist/ACC0000031/sections/impression...complete

readRead `/worklist/ACC0000031/sections/history.md`complete

readRead `/worklist/ACC0000031/sections/technique.m...complete

readRead `/worklist/ACC0000031/sections/comparison...complete

flagraise_flagcomplete

1 flag raised

Ask the agent... (/ opens the commands · Enter to send, Shift+Enter for newline)

Send

Right UPJ calculus in FINDINGS, *left* in IMPRESSION. The agent raises a **flag** and marks the line — it changes nothing.

ACP-Rad Demo · technology demonstration — not for clinical use

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The sign-off gate

ACP-Rad

EMERGENCY MDCT OF THE BRAIN · ACC0000001

resident

attending

2 flags

draft

Sign off

research_synthetic

QA unavailable — the check was stopped.

B I

☞ Commands

EMERGENCY MDCT OF THE BRAIN

HISTORY: Known case of hypertension and atrial fibrillation presents with acute right hemiparesis and aphasia for 2 hours. Physical examination reveals GCS of E4V2M6 and motor power grade 2/5 on the right side. The CT scan was requested to rule out intracranial hemorrhage.

TECHNIQUES: Axial helical scan of the brain was performed using 2.5-mm slice thickness with coronal and sagittal reconstructions.

Intravenous contrast brand, concentration, volume: Not used.

Immediate complication: None.

Estimated radiation dose: 58 mGy, 890 mGycm. Standard DRL recommended by the American College of Radiology and European Commission = 75 mGy, 1050 mGycm. Values reported for 16-32-cm PMMA phantom. Uncertainties associated with this method are approximately 20%.

COMPARISON: None.

FINDINGS:

Cerebral parenchyma: Ill-defined hypodense lesion involving the left insular cortex, left lentiform nucleus, and left frontal operculum with loss of gray-white differentiation. No intraparenchymal hemorrhage.

A well-defined hypodense lesion at the right thalamus, probably chronic lacunar infarction.

Patchy hypodense areas involving periventricular and deep white matter of bilateral fronto-parietal lobes, likely ischemic white matter change related to small vessel disease.

Cerebellum and posterior fossa: Normal given a limited evaluation of the brainstem due to beam-hardening artifacts. The cerebellum appears unremarkable.

Extraaxial spaces: No extra-axial collection is observed.

Ventricles: Normal size of the ventricular system. No intraventricular hemorrhage.

Midline shift: None.

Brain herniation: None.

Vascular system: Hyperdense left M1 segment of the middle cerebral artery. Atherosclerotic changes of bilateral intracranial ICAs and vertebrbasilar system.

Calvarium and scalp: No fracture identified.

Skull base, sella and temporomandibular joints (TMJs): Normal.

Visualized orbits, paranasal sinuses and mastoid air cells: Mild mucosal thickening of bilateral ethmoid sinuses. The rest of visualized both orbits, paranasal sinuses, mastoid air cells, and nasopharynx appear unremarkable.

Visualized upper cervical spine: No fracture identified.

IMPRESSION:

• ...

rad-report-agent

L2

openai:gpt-5.6-terra

ready

Chat

Audit · 8

critical uncommunicated

sections/findings.md · Line 10

FINDINGS reports a hyperdense left M1 segment with acute left MCA-territory ischemic change (lines 10 and 2), but the report records no communication of this critical acute stroke finding.

Acknowledge

omission

sections/findings.md · line 2

FINDINGS describes acute left MCA-territory ischemic

/qa

search

ls

complete

search

ls

complete

read

Read `./worklist/ACC0000001/report.md`

complete

search

ls

complete

read

Read `./worklist/ACC0000001/sections/findings.md`

complete

read

Read `./worklist/ACC0000001/sections/impression...`

complete

flag

raise_flag

complete

flag

raise_flag

complete

stopped

Ask the agent... (/ opens the commands · Enter to send, Shift+Enter for newline)

Send

Prelim / Sign off run `/qa` first. An acute stroke with no *discussed with* line is flagged *critical, uncommunicated*. *Sign off anyway* stays your call.

Ask anything

ACP-RadEMERGENCY MDCT OF THE WHOLE ABDOMEN · ACC0000031residentattending

B I

☞ Commands

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The scan phases include: plain, late arterial, venous, 3-minutes delayed.

Intravenous contrast brand, concentration, volume: Iohexol 350 mgI/ml; 80 ml.

Other route(s) of contrast: None

Immediate complication: None.

Estimated radiation dose: 14 mGy, 690 mGycm.

COMPARISON: None.

FINDINGS:

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IMPRESSION:

- A 6-mm left ureteropelvic junction stone.
- A hepatic cyst.

1 flagdraftSign offresearch_synthetic

rad-report-agentL2openai:gpt-5.6-terra

ready

ChatAudit · 4

discrepancysections/impression.md · line 2

The IMPRESSION identifies a left UPJ stone, whereas FINDINGS describes a right UPJ calculus with mild right hydronephrosis and perinephric fat stranding.

Acknowledge

What can I improve the writing of the impression?

searchlscomplete

read

Read ` /worklist/ACC0000031/sections/impression...

complete

read

Read ` /worklist/ACC0000031/sections/findings.md`

complete

flagraise_flagcomplete

The key improvement is to include the obstructive sequelae and correct the laterality to match
Kidneys and ureters: in the findings. I also recommend specifying the cyst as simple and including its size/location only if clinically useful.

Suggested wording:

IMPRESSION:

- 6 mm right ureteropelvic junction calculus with mild right hydronephrosis and perinephric fat stranding.

- 1 cm simple hepatic cyst in segment VI.

I flagged the right/left laterality discrepancy for review.

Ask the agent... (/ opens the commands · Enter to send, Shift+Enter for newline)

Send

Free text works too. If the answer wants to edit the report, it goes through the same gate as a skill.

ACP-Rad Demo · technology demonstration — not for clinical use

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Everything is on the record



One /impression, read bottom-up:

RECORD	MEANING
session.new → manifest=19	the agent was told which 19 paths it may read
fs.read sections/findings.md	what it looked at
proposal.received → 1 hunks	what it wanted to change
permission.request	the editor asked you
hunk.accept_edit	your decision — <i>Accept for review</i>
fs.write.applied	what actually landed

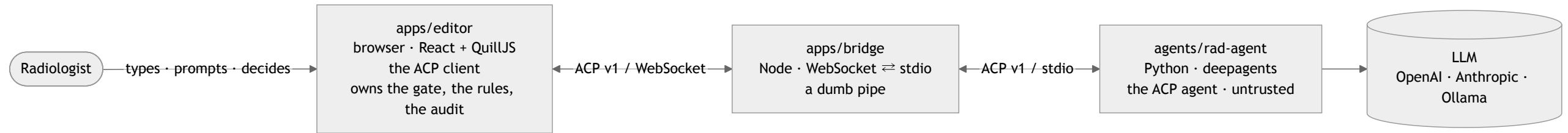
Same records in the sidebar and in `audit/{accession}.jsonl` on disk.

When a colleague asks “what did the AI do to this report?”, this is the answer.

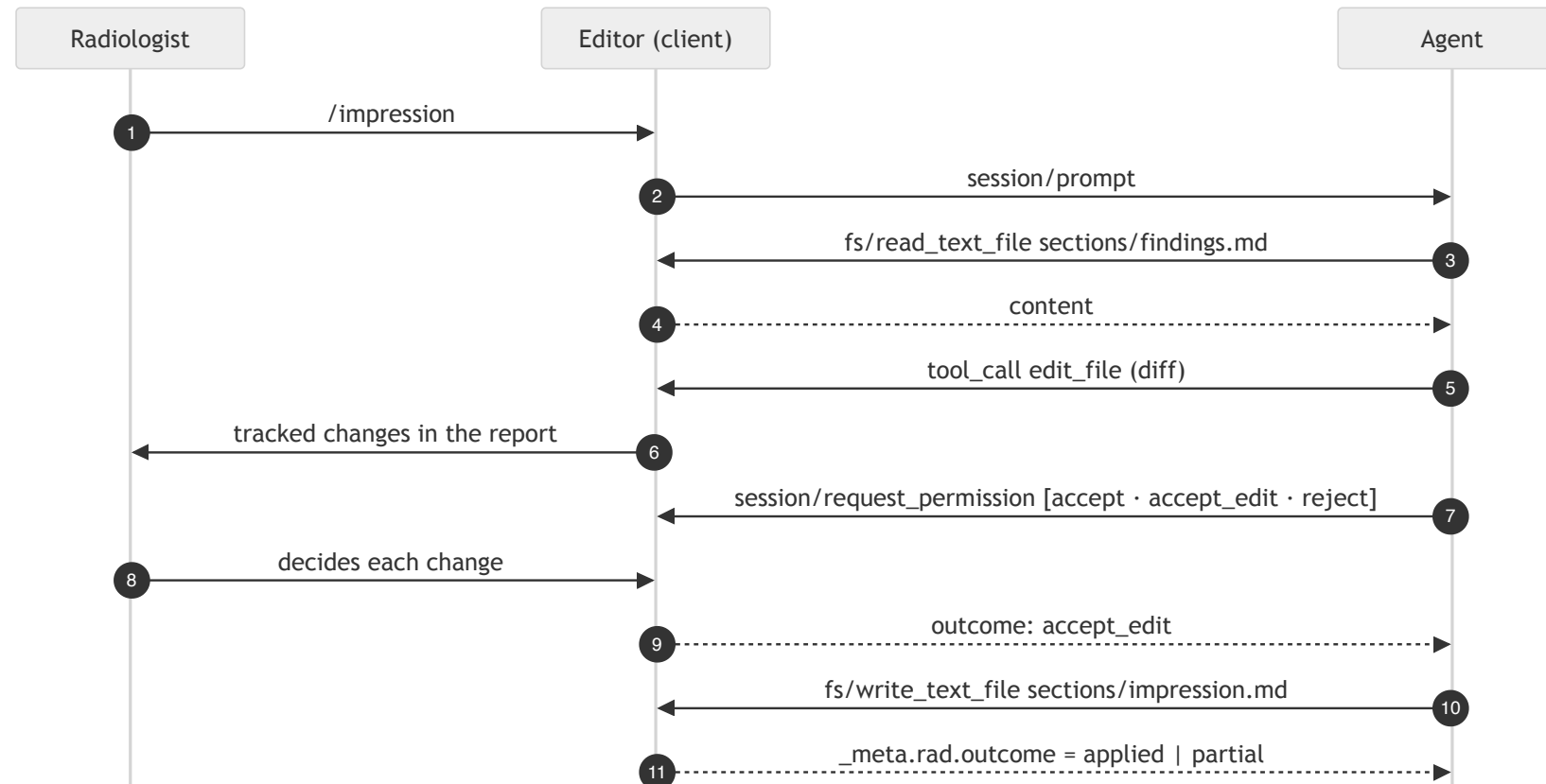
Under the hood

Four slides for the engineers. Skip freely.

Three boxes



The browser *is* the protocol client. The bridge only exists because agents speak stdio. Anything that enforces a rule lives in the editor.



Standard ACP calls; radiology rides in `_meta`. The agent's write is *acknowledged*, never applied — the accepted text is already in the buffer.

ADDITION	WHERE IT RIDES	WHY
Namespace + manifest — sections, metadata, priors, templates, snippets as paths	<code>session/new _meta.rad</code>	the agent needs a map of what it may read
Clinical verbs <code>accept</code> · <code>accept_edit</code> · <code>reject</code>	<code>request_permission</code> options	decisions mean something in the audit; never “always allow”
Write outcome <code>applied / partial</code>	<code>fs/write_text_file _meta.rad</code>	the agent learns what actually landed
<code>_rad/flag</code>	custom method	QA without an edit
Level 0 / 1 / 2	<code>initialize _meta.rad</code>	graceful degradation: a plain ACP agent still works

Only sanctioned extension points — `_meta` and `_`-prefixed methods. A conforming ACP peer never sees a field it must reject.

$M \times N \rightarrow M + N$, honestly

Demonstrated — by construction

- The editor has no agent-specific code path.
- The agent has no editor-specific code path.
- Swapping is safe because the invariants sit on the client.

Not yet — by measurement

- $N = 1$ live agent; two off-the-shelf agents are wired, not exercised.
- $M = 1$ editor; a headless test client is weak evidence.
- Every real client grows a small quirks layer per agent — LSP did too.

It becomes a *result* when a foreign editor drives this agent, and a foreign agent completes a scenario in this editor, both unchanged.

What next

What this is, what it isn't, and where the profile goes.

What this is — and isn't

Is

- A working prototype: template → draft → compare → QA → sign-off gate
- One editor, one protocol, any agent
- The human gate, enforced by the editor, auditable

Isn't

- A product, or a medical device
- Validated on real reports — all data synthetic
- Tied to one model — switchable at runtime

For you, the radiologist: what would you want the agent to do — and what must it never do?

Try it

```
git clone https://github.com/radrama-ai/acp-rad-demo.git
cd acp-rad-demo && just install && just dev
```

- Runs on a laptop; hosted or local (Ollama) models.
- Design, protocol shape and the M×N analysis are in docs/.
- The profile draft: docs/ideas/acp-rad-protocol-proposal.md → a standard, next.

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